

Ritesh Jat

B.Tech in Electronics and Communication Engineering | Maulana Azad National Institute of Technology, Bhopal

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Education

Maulana Azad National Institute of Technology, Bhopal

2023 – Present

B.Tech in Electronics and Communication Engineering

CGPA: 8.59

Govt. Excellence H.S. School, Sheopur (M.P.)

2022

Class XII – MP Board

96.7%

Alpha H.S. School, Sheopur (M.P.)

2020

Class X – MP Board

99.3%

Technical Skills

Languages	C, C + +, Python, SQL, JavaScript, TypeScript
Web Development	HTML, CSS, Tailwind CSS, React.js, Next.js, Node.js, Express.js, REST APIs
Databases	MongoDB Atlas, SQL
Libraries / Tools	Pandas, NumPy, Matplotlib, Streamlit, Librosa, Git, GitHub
Core Areas	Data Structures & Algorithms, DBMS, OOPs, Computer Networks, Operating System
Technologies	RAG Systems, JWT Authentication, Vercel, Render, Automation, IoT Fundamentals

Leadership & Experience

MANIT Bhopal

2025 – Present

Student Mentor

Bhopal, M.P.

- Selected among 30 mentors to guide 6 first-year students in academics, coding practices, and career development.
- Assisted mentees in adapting to college life and improving problem-solving and analytical skills.

PrepRight

Summer 2025

IoT Summer Intern

Remote

- Completed a remote internship focused on IoT concepts, sensors, and embedded systems applications.
- Gained practical exposure to microcontrollers and real-world IoT system workflows.

MANIT Bhopal

2023

Mess Committee Member

Bhopal, M.P.

- Coordinated mess operations and communication for facilities serving 700+ students, ensuring smooth functioning.

Projects

Full-Stack Portfolio CMS Platform

Next.js | TypeScript | Node.js | MongoDB

GitHub Live

- Developed a full-stack portfolio CMS with Next.js, Express.js, MongoDB Atlas, and TypeScript for managing projects, blogs, and contact messages.
- Implemented JWT-based authentication, protected admin routes, and secure CRUD operations through a dedicated admin dashboard.
- Deployed the frontend on Vercel and backend on Render, ensuring responsive design and seamless API integration.

AI-Based Teaching Assistant (RAG System)

Python | NLP | LLMs

GitHub Live

- Built a Retrieval-Augmented Generation (RAG) assistant capable of answering queries from custom documents.
- Implemented embeddings and similarity search for context-aware response retrieval.
- Integrated LLM-based answer generation to improve response accuracy and relevance.

Criminal Face Generation & Recognition Web App

Python | React | Flask | ML

GitHub Live

- Developing a web-based facial composite generation and criminal recognition system.
- Applied computer vision and ML techniques for feature-based facial identification.
- Designed an interactive interface for assembling and matching facial components.

Advanced Speaker Recognition System

Python | DSP | MFCC | Streamlit

GitHub Live

- Developed a DSP-based speaker recognition system utilizing Mel-Frequency Cepstral Coefficients (MFCC) for feature extraction and cosine similarity analysis.
- Implemented real-time audio recording, spectral visualizations, and a comparison dashboard using Streamlit, Librosa, and NumPy.

Achievements

- Solved 450+ DSA problems across LeetCode and GfG, strengthening problem-solving and coding proficiency.
- Ranked 5th in Class X and 9th in Class XII in MP Board examinations with consistent academic excellence.
- Qualified JEE Advanced 2023 with an All India Rank (AIR) of 13,486, securing admission to MANIT Bhopal.